

EDUCATION

- **MIT** Cambridge, MA
Undergraduate, Computer Science and Mathematics (Double Major) Sep 2024 – Present
Selected Coursework: Deep Generative Models; Quantitative Methods for NLP; Representation, Inference and Reasoning in AI; Quantum Computation GPA:5.0/5.0
- **Tsinghua University** Beijing, China
Preparatory Program, Institute of Interdisciplinary Information Science (IIIS, Yao Class) Sep 2023 – Jun 2024.
Selected Coursework: Advanced Calculus; Linear Algebra; Abstract Algebra; Algorithm Design; Introduction to Computer Systems; Introduction to LLM Applications; Deep Learning; Exploration to Scientific Research of Lab; GPA:4.00/4.00

SKILLS

- **Programming:** Python(Jax/Pytorch for Deep Learning Models), C, C++, Lean
- **Language:** English(proficient), Chinese(native)

EXPERIENCE

- **Internship in Agentic Pipeline** Abaka AI
Research Engineer Jul 2026 - Sep 2026
Internship at Abaka AI doing agentic system to improve data pipeline, and doing some independent research.
- **Internship in Automatic Theorem Prover** ByteDance Seed
ByteDance AI4Math Team Jul 2025 - Dec 2025
Internship on automatic theorem provers using AI methods. Developed the algorithm for Seed-Prover, which is an automatic agentic system for proving olympiad-level math problems. It achieved a silver medal in IMO2025 and had the SOTA performance on PutnamBench and other hard math datasets.
- **Undergraduate Research in Deep Generative Models** MIT
Supervised by Kaiming He Sep 2024 -Present
Research on deep generative models in computer vision, especially diffusion models and flow matching models, and investigate the underlying principles of denoising-based models, and develop new models on TPU machines using Jax.
- **Undergraduate Research in Technical Computer Science** Tsinghua University
Supervised by William Kuszmaul Feb 2024 - Sep 2024
Research on the theoretical analysis of randomized data structures. Developed a new time complexity bound and design an algorithm for a general framework of a randomized problem.

AWARDS

- **Top 25** 2025
The William Lowell Putnam Mathematical Competition
- **Gold Medal** 2023
64th International Mathematics Olympiad(IMO)
- **First Prize** 2021
National Olympiad in Informatics in Provinces(NOIP)

PROJECTS

- **Video Style Transfer(Deep Learning Course Project)** Feb 2024 - June 2024
Python, Deep Learning
An automatic pipeline to do image style transfer by user-specified prompts using ControlNet and frame interpolation.
- **Speeding Up Diffusion Models with One-step Generators** Sep 2024 - Dec 2024
Python, Deep Learning, Generative Models, Diffusion Models, VAE
In the project, we proposed a new method to speed up the training of diffusion models by using one-step generators. On toy experiments, this reduces NFE by half while maintaining the sample quality. We also wrote a blog post, explaining the motivation of the experiment from a higher perspective.

Feb 2024 - Jun 2024

- **Knowledge Database**

Python, LLM, Bash, Makefile

In the project, we apply LLMs to answer user questions given a folder containing documents as the context. We developed a tagging system, which make the search efficient even when the number of documents is large. We also support semantic search for multimodal documents, such as images and videos.

Apr 2025 - Present

- **Auto GCP TPU Management**

Python, TPU, GCP, Job Management

An slurm-like automatic job manager for Google Cloud Computing(GCP) development in TPUs serving as the infra of the research group. Including automatic environment solver, reapplying resuming jobs that use a preemptible TPU which has been preempted, etc.

PUBLICATIONS

- **Is Noise Conditioning Necessary for Denoising Generative Models?**

ICML 2025

Zhicheng Jiang, Qiao Sun*, Hanhong Zhao*, Kaiming He*

- **Seed-Prover: Deep and Broad Reasoning for Automated Theorem Proving**

ByteDance Seed AI4Math

- **Seed-Prover 1.5: Mastering Undergraduate-Level Theorem Proving via Learning from Experience**

ByteDance Seed AI4Math

- **Bidirectional Normalizing Flow: From Data to Noise and Back**

Yiyang Lu, Qiao Sun, Xianbang Wang, Zhicheng Jiang, Hanhong Zhao, Kaiming He

- **One-step Latent-free Image Generation with Pixel Mean Flows**

Yiyang Lu, Susie Lu, Qiao Sun, Hanhong Zhao, Zhicheng Jiang, Xianbang Wang, Tianhong Li, Zhengyang Geng, Kaiming He